

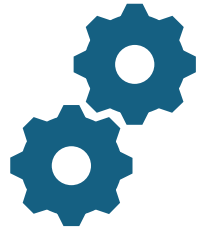


upGrad Led Foundation Bridge

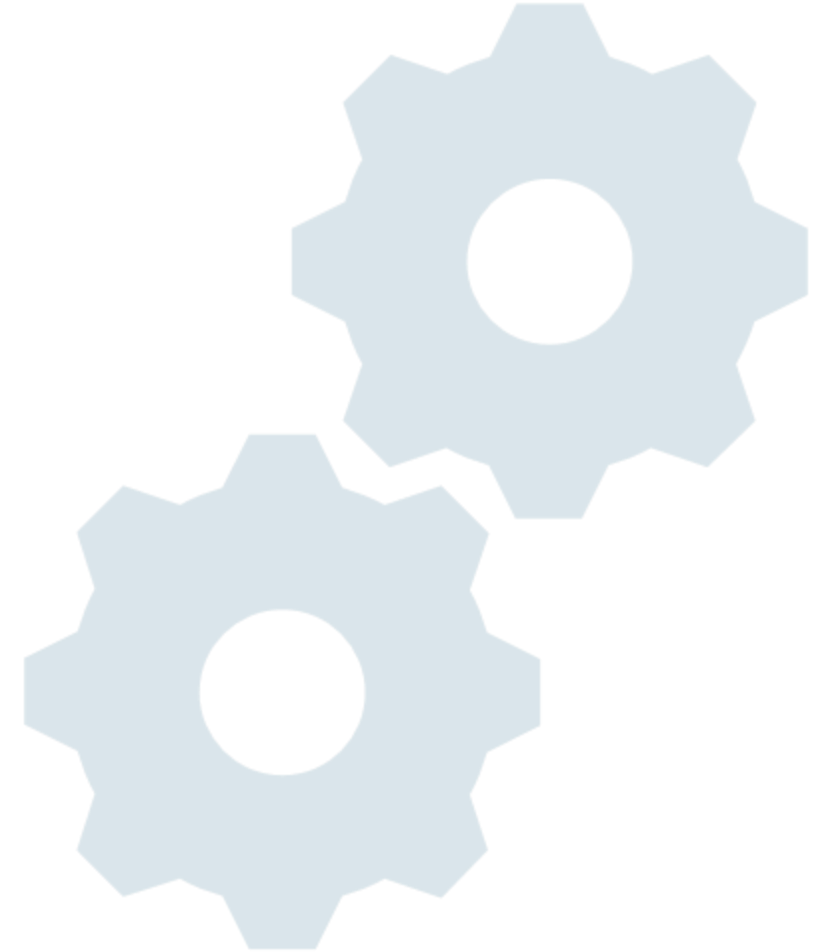


Sessions

- AI Engineering Environment Setup
- Mathematical Foundations for AI Systems
- Python Engineering for AI Workflows
- Data Engineering for AI Applications
- Basic ML – I
- Basic ML – II
- DL Basics
- NLP Basics



AI Engineering Environment Setup



Spread

- AI Engineering Environment Fundamentals
- Python & Virtual Environment Setup
- IDE, Jupyter and VS Code Setup
- GPU, CUDA and Runtime Awareness
- Cloud Notebooks and GPU Platforms
- Installing GenAI Libraries
- APIs, Keys and Endpoint Setup
- Running First LLM Application
- Troubleshooting & Best Practices

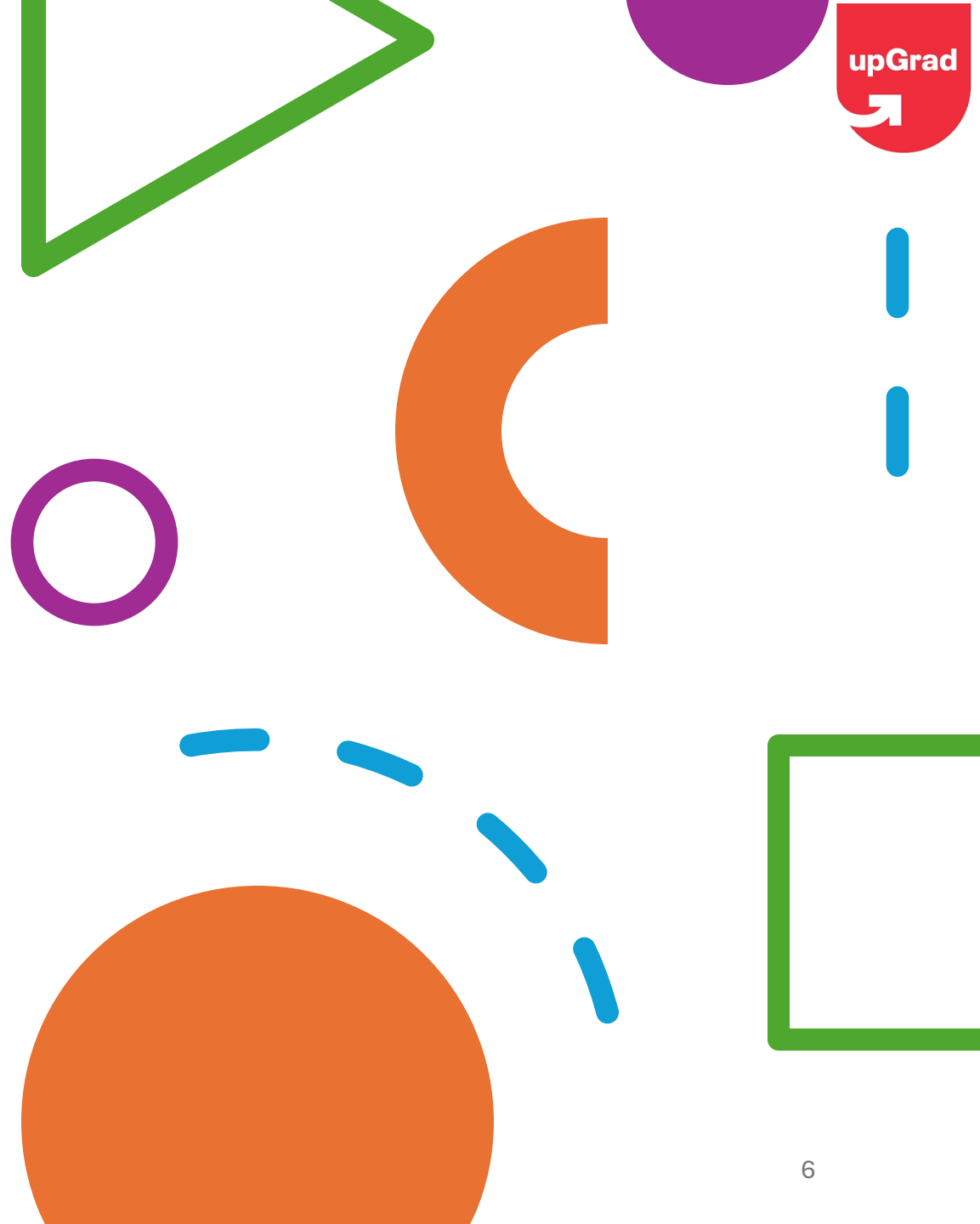


AI Engineering Environment Fundamentals

- What is AI Engineering Setup
- Why GenAI Environments are different
- CPU vs GPU
- Local vs Cloud Execution
- Why dependency management matters
- Typical failures in GenAI projects

Python and Virtual Environment Setup

- Python Installation
- Pip basics
- Virtual environments
- Package Isolation
- Dependency conflicts
- Docker Introduction



IDE, Jupyter and VS Code Setup

- Why IDE Matters
- Jupyter notebook workflow
- VS Code Setup
- Python extensions
- Notebook execution

GPU, CUDA and Runtime Awareness

- What is GPU acceleration ?
- VRAM vs RAM
- CUDA basics
- GPU usage in Google COLAB
- Why LLMs need GPUs

Cloud Notebook & GPU Platforms

- Why cloud platforms are used ?
- Free GPU options
- COLAB basics
- Runtime Selection
- Connecting Google Drive

Installing GenAI libraries

- Transformer ecosystem
- PyTorch basics
- Installing AI libraries
- Version Compatibility

APIs, Keys & Endpoint Setup

- What are API Keys ?
- Endpoint basics
- Hosted LLMs
- Authentication
- Environment variables

Running First LLM Application

- Loading transformer models
- Running inference
- Prompt based generation
- Basic text generation workflow

Troubleshooting and Best Practices

- CUDA mismatch
- Package conflicts
- OOM errors
- Broken kernels
- Dependency troubleshooting
- Notebook restart practices

Thank you 😊

